Astrophysical Recipes
The Art of AMUSE

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In memoriam

In memoriam Inti Pelupessy
Gallo 1977 - Leiden 2024

This photograph shows Inti Pelupessy at the Archeon Museum Park in the Netherlands, trying to make fire by rubbing together two wooden sticks. Inti was strongly involved from the start in the AMUSE project. He passed away in the summer of 2024, and is missed by all.

As a brilliant hydrodynamics expert (trained by the very best), Inti quickly became a key figure in the development of AMUSE, his swift mastery of this complex framework is a testament to his remarkable intellect. His pioneering work in this area resulted in important contributions to the field. Without his outstanding qualities as a researcher, his sharp thinking, his excellent understanding of the physics and his unsurpassable programming skills, AMUSE would simply not exist.

Possibly even more striking were his social skills. As you can see on the photograph, Inti was a natural educator, teaching on subjects ranging from procedural operations to thermodynamics, here demonstrated in the making of fire. Inti was reluctant to promote his own work, and always listened to the opinions of others before elaborating on his own ideas. Still, he was often one of the strongest minds in a collaboration.

Those of us fortunate enough to have known Inti will remember his deep insight, kindness, patience, and his dedication to science. His loss is felt deeply by us, and by all in his periphery but his legacy in both science and mentorship will live on.

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Preface

This is the second, revised, edition of the academic textbook on the Astrophysics Multipurpose Software Environment, or AMUSE, for short (Portegies Zwart & McMillan, 2018). The AMUSE project started at the MODEST-5 conference in Hamilton, Ontario, in August 2004. Steve McMillan and Simon Portegies Zwart were sipping coffee, nibbling on cookies, and discussing the future of `starlab`.

`Starlab` began in the early 1990s as an alternative and competitor to Sverre Aarseth's NBODY series of codes. In collaboration with Piet Hut and Jun Makino, we wrote a modular package combining stellar dynamics and evolution using modern programming paradigms and data structures. The code was written in C++, an up-and-coming language at the time, which afforded many opportunities to go beyond the rigid arrays used in existing languages at the time.

The `starlab` package contained many novel features not found in comparable state-of-the-art codes even today. Most codes at the time, and certainly the leading NBODY series developed by Sverre Aarseth, were "monolithic" solvers, designed, written, and integrated for one particular purpose. A direct summation N-body code with a hard-coded realization of stellar evolution is an excellent example of a monolithic code. Because monolithic codes are written in a single programming language with simple data structures, they are generally relatively uncomplicated, straightforward to compile, and quite portable.

By contrast, `starlab` was a non-monolithic code composed of two independent components; a gravitational N-body solver `kira` and the binary population synthesis code `SeBa`. Much of the internal coding in `starlab` was devoted to maintaining consistent communication between its constituent parts. The result was still quite tightly coupled, but it preserved the essential kernel of independent communicating parts. But as we contemplated expanding our scope by including hydrodynamics, we realized that, although we had anticipated this possible expansion, we still did not really know how to do it without radically changing the basic data structures and rewriting the code. Hard-coding a hydro module into `starlab` would be counter to all our design choices and would be horribly complicated to code and maintain, but how else might we expand the code's functionality?

After more than a decade of reliable service and hundreds of scientific papers, `starlab`, we realized, faced complicated design choices for the future expansion of the package. Or should we start all over again?

We had no clear solution to the problem, and we felt insufficiently confident to start all over until Sverre Aarseth and Peter Eggleton joined us. We started talking about the codes they had written. Sverre initiated and led the NBODY series of direct

summation N-body solvers. His codes are highly optimized and always written in a historic FORTRAN dialect. His latest incarnation NBODY6, at that time, was the old-school counterpart of `starlab`. NBODY6 was faster than `starlab`, but (we think) was less elegantly structured, if one can speak of any formal structure. Peter is a pioneer in writing stellar structure and evolution codes. The time Sverre spent writing N-body codes, Peter spent improving `EVTwin`, his stellar evolution code, also in vintage FORTRAN.

Sverre looked very happy to hear that we considered `starlab` past its prime. "NBODY6 is brand new" he told us, "very alive, and kicking bodies, yes." We thought that, in some way, we might benefit from the many decades of combined experience of these two grandfathers of computational astrophysics. Why redesign code? Why rewrite any software if we already have the right tools at hand? If only they would turn into useful tools with a proper interface. Could it be possible to incorporate one of these codes into the other? We asked Sverre how he would feel about this. He responded: "Wouldn't it be nice to have Peter's stellar evolution code as a part of my beautiful N-body code?" Peter responded: "But, Sverre, my dear friend, how splendid would it be to have your N-body code as part of my stellar evolution code."

That got us thinking. Maybe we should get away from "mine" and "thine," and operate instead in a collective framework, and we proposed: "What about if both your codes could work together?" While hardly ever in agreement, both men thought this would be an interesting idea, like the Borg (of whom we suspect neither had ever heard) assimilating the positive characteristics of everybody to become better as a whole. But they were even more sure that such an attempt would be impossible. You would have codes written in all sorts of languages. How would you deal with all the nasty communications and unit conversions worldwide? Also the communication between codes will become a major bottleneck. No, better to have another cookie and dream about NBODY7.

By now, NBODY7 turned into NBODY6++; still in FORTRAN, though. But in that conversation, we had the kernel needed to get started; together, we can rule the Galaxy. That evening, we had a quiet dinner and discussed the possibilities of a language-independent framework in which codes would operate as modules that carry out specific tasks. The communication between the various codes would have to go through some sort of interface, to be determined, allowing it to perform operations on the data, formatting it to the requirements of the other codes, and scheduling the execution of those codes. Maybe it was not so impossible after all.

The MODEST initiative, which gave rise to the Hamilton conference where this exchange took place, was a direct outgrowth of our experience with `starlab`. However, it eventually became clear that, even with the best intentions, we still had combined algorithms in a way that made it hard to make radical changes to the fundamentals.

This realization led Piet Hut and Mike Shara in 2001 to organize a conference at the American Museum of Natural History in New York, to bring together experts in stellar dynamics, stellar evolution, stellar hydrodynamics, computer science, and other relevant areas, to seek truly modular ways to combine codes to create a new computational framework. The meeting was subsequently named MODEST-1 (for MOdeling DEnse STellar systems). Perhaps unsurprisingly, the early sample codes that emerged from

the meeting looked very much like `starlab`, but MODEST (subsequently re-acronymed as Modeling and Observing DEnse STellar systems) rapidly became an important forum for the discussion of all aspects of dynamical modeling and observations of dense stellar systems.

Early MODEST meetings focused more on astrophysics than computation. Still, following the MODEST-5 meeting in Hamilton, a rapid series of satellite meetings changed the software landscape. We organized a meeting in Lund, Sweden, in December 2005 and formulated a concrete plan for combining existing codes. The discussion continued a few months later in Amsterdam, where we organized another small workshop to discuss the possibilities. Piet Hut was there, along with Jun Makino, Peter Teuben, Pierro Spinnato, and Breanndán Ó Nualláin. Breanndán was a Python aficionado, and with his help, we had the first version of a rudimentary working code operational in a few days. We called this framework MUSE, for Multipurpose Software Environment.

The key idea of MUSE was that no code got to "drive" the simulation. Dynamical, stellar evolution and other codes were invoked as modules, with suitably defined interfaces, but the top-level code was a Python script that created and coordinated the scientific study. The implementation involved heavy use of Python tools such as `f2py` and `swig`, and ran into numerous practical problems that needn't concern us here, but the basic idea of a script-driven, democratic code framework was established.

After the fundamental ideas had been put into practice, and once some of their shortcomings were recognized, we realized that a bunch of astronomers would not be able to make quick progress on such a large software environment. We searched for funding to hire a dedicated software architect, Arjen van Elteren, who could help design our framework, and two computational astronomers, Inti Pelupessy and Nathan de Vries, to develop the first science cases with the new framework. The team turned out to be very effective. The framework is now called AMUSE, for Astrophysical MUltipurpose Software Environment. Over the last decade we worked hard with students to improve the package, broaden its scope and widen the scientific applications. The most recent developments include incorporating codes for simulating exo-planetary atmospheres and interstellar chemistry, and the entire build-system was renewed. This manuscript explains some of the functionality and operations possible with the framework.

The main objective of this book and the entire AMUSE framework is to explore new science. To seek out new phenomena and new discoveries. To boldly go where no computer has gone before! This is the continuing mission of this exciting journey. The first edition of the AMUSE book, published in 2018, reflected the authors' roots in simulations of dense stellar systems, as well as the historical development of the field. In this new version, we sought new collaborator and found him in Steven Rieder. He was closely connected to the AMUSE endeavor from the beginning; somewhat junior at the time of the initial development, he has developed to a true AMUSE master, and took care of most of the maintenance, documentation, updates, bug fixes, et cetera; AMUSE's presence is strong in our academic family[1].

[1] Adopted from Luke Skywalker to Leia Organa in Star Wars: Return of the Jedi (1983).

The new book is divided into three parts. In part I, we discuss the fundamentals of AMUSE and the working of the monolithic solvers; gravity, stellar evolution, hydrodynamics, and radiative transfer. In part II, we discuss elementary coupling strategies adopted in AMUSE. Part III brings all AMUSE capabilities together and shows the reader how to design complicated multi-scale and multi-physics experiments. In the last part, we also show how AMUSE can be adapted and applied to an even broader range of problems. We present complete AMUSE scripts for performing high-fidelity simulations and provide some examples outside the domain of astrophysics.

But above all, the goal of this book is to have fun. To enjoy computational astrophysics, and to explore the nonlinear couplings between different physical domains without having to worry too much about the details of energy conservation in a close three-body interaction or managing the arcana of high-dimensional differential equations. Decades of computational astrophysics research have gone into some of the modules incorporated into AMUSE. These are trustworthy codes and form an excellent basis for further study. Still, never trust a code that doesn't work, but trust a working code even less.

The AMUSE framework is still a work in progress. Even after more than a decade of coding, thinking, and writing, we suspect this book and the framework to contain errors and typos that neither we, our reviewers, nor our editors caught. For these reasons, we will maintain an evolving document in AMUSE that lists corrections to the published text and highlights developments in AMUSE that may interest readers, such as new algorithms, functionality, or insights into the many problems we address. It can be found in `${AMUSE_DIR}/examples/textbook/updates/AMUSE_updates.pdf`, which can be saved via

```
from amuse.examples import export
export("AMUSE_updates.pdf", savedir="./")
```

where `amuse.examples` is the examples package companion to this book (see Section 1.4). Incidentally, the "laptop" cited in the timing estimates for most figures is Simon's Lenovo P14s, 20-core 3th Gen Intel Core i7-1370P with NVIDIA RTX A500 GPU. Your timing may differ.

Simon Portegies Zwart
Steve McMillan
Steven Rieder

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Acknowledgments

> I don't know half of you half as well as I should like; and I like less than half of you half as well as you deserve.
>
> J.R.R. Tolkien

These words were uttered by Bilbo Baggins on 22 September 3001 in front of 144 invited guests to celebrate his eleventy-first birthday. With this speech he wanted to say farewell to and acknowledge friends and family. We also would like to acknowledge a large number of people who contributed to this book, although we will not go so far as to insult half of them. We at least do not consider any of them at the same level as Otho and Lobelia.

AMUSE is really a community effort, and a large number of people have contributed. The master students at Leiden Observatory helped eliminating bugs and typos during the various courses on computational astrophysics. We apologize to them for having used them as guinea pigs for the first version of the manuscript[2]. We particularly thank the many people who contributed by adding new code, developing or debugging scripts, or initiating new ideas through stimulating discussions. They include: Assilkan Adilkhan, Eric Andersson, Sabrina Appel, Jeroen Bédorf, Tjarda Boekholt, Anthony Brown, Maxwell Cai, Claude Cournoyer-Cloutier, Niels Drost, Will Farner, Michiko Fujii, Evghenii Gaburov, Joseph Glaser, Guilherme Goncalves, Adrian Hamers, Erwan Hochart, Shuo Huang, Piet Hut, Masaki Iwasawa, Lucie Jílkova, Nadia Kempers, Rony Keppens, Ralph Klessen Rob Knop, Jun Makino, Tunde Meyer, Sean Lewis, Carmen Martínez-Barbosa, Mordecai-Mark Mac Low, Breanndán Ó'Nualláin, Jan-Pieter Paardekooper, Veronica Saz Ulibarrena, Alison Sills, Jelmer Stroo, Andrew Pellegrino, Brooke Polak Silvia Toonen, Aaron Tran, Alessandro Trani, Gijs Vermariën. Serena Viti, Joshua Wall. Long Wang, Julia Wasala, Maite Wilhelm, Alfred Whitehead, and Thomas Wijnen.

Several of the nicest figures were made by Edwin van der Helm (Figure 8.5), Alex Rimoldi (Figure 6.4), Nora Lützgendorf (Figure 10.13), and Sabria Apple Figure 10.4.

We received illuminating private lectures on gravitational dynamics (Chapter 4) by Sverre Aarseth, on stellar evolution (Chapter 5) by Onno Pols and Peter Eggleton, on hydrodynamics (Chapter 6) by James Lombardi and Volker Springel, and on radiative transfer (Chapter 7) Barbara Ercolano and Kees Dullemond.

During the development of AMUSE and the writing of the book we made extensive use of many additional public facilities and packages, including: `ADS`, `arXiv`, `CDS`, `git`,

[2]Here we refer to the Cavia porcellus, in honor of the 1901 Nobel prize in Medicine, awarded to E. Adolf von Behring, who used this cute furry rodent to develop a serum therapy against diphtheria.

`hdf5`, `ipython`, `Jupyter`, LaTeX, `linux`, `MPI`, `matplotlib`, `numpy`, `OpenMP`. `python`, `Seaborn`, `wikipedia`, and probably many more we forgot to mention.

Both authors had the opportunity to spend an extensive amount of time on this book, for which we are grateful to our (former) host institutions, Leiden Observatory, Drexel University, Geneva Observatory, KU Leuven and the Anton Pannekoek Institute for Astronomy. But we are also grateful to the Center for Planetary Science in Kobe, the Lorentz Center in Leiden, Geneva Observatory, the University of Strasbourg, the Anton Pannekoek Institute, Universiteit Leuven for hosting our discussions. Parts of the development of AMUSE were financed by NOVA NWO (639.073.803 and 614.061.608), NASA (grants NNX07AG95G, NNX08AH15G), NSF (AST0708299), H2020 FET CompPat, the EU P7/08 CHARM project and the NLeSc. We used compious amounts of Supercomputer time on SURF/Snellius (2023.008, 2024.058, EINF-10097, EINF-11229, EINF-12718, EINF-13923) and LUMI (EINF-11368).

Without Arjen van Elteren, Inti Pelupessy, and Nathan de Vries, AMUSE would not exist, and we would never have been able to write this book. A special thank you also is due to the anonymous referee Douglas Heggie (of the first version), who identified many omissions and errors, which we hope we have corrected. We also thank our editor and staff at IoP and in particular Robbie Trevellian, Leigh Jenkins, and Daniel Heatley for their support of this writing project.

Our spouses and children earn considerable praise for their patience. They are very happy that this is over, and we are happy that they are still there.